NESTED-SPACE COSMOLOGY

The Geometry of a Universe

Regions, boundaries and what an interior observer can learn

AUTHOR
DOUGLAS EK

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Abstract

Nested-Space Cosmology (NSC) investigates whether one inherited geometric law can connect nested regions, their exchange of energy across event horizons, and the quantities measured by an interior observer. The foundation rests on five openly declared geometric axioms (**A0--A4**): (**A0**) geometry follows the path of least resistance by minimizing the 1D Cauchy--Schwarz generator stretch cost plus enclosing compression work, $E = Q_{stretch} + PV$, without fluid incompressibility or borrowed surface tension; (**A1**) a 1D spatial seed plus an inherited compression offset (time) generates a closed 2D circle and 3D sphere; (**A2**) seen edge-on, the closed circle $z(\tau)=R_0 e^{i\omega\tau}$ is the harmonic pendulum $R(\tau)=R_0\cos(\omega\tau)$, which crosses `0` smoothly into the negative direction $R<0$ (`-`), flipping the 3D orientation Jacobian $\det F=(R/R_0)^3$ from `+` to `-` and turning inward implosion ($\dot{R}<0$) into outward child expansion ($\dot{r}_-=-\dot{R}>0$) without the artificial velocity cusp $\Delta\dot{r}=+2R_0\omega$ created when standard physics restricts radius to an unsigned size $r=|R|\geq 0$; (**A3**) each nested room carries a moving wavelength band $[\lambda_{min}(\chi),\lambda_{max}(\chi)]=e^{-\chi}[\lambda_{min,0},\lambda_{max,0}]$ bounded by its Planck cutoff (λ_{min}) and Absolute-Zero horizon (λ_{max}); and (**A4**) every room sits between an outer birth `0` valve and inner black-hole `0` valves on a closed vortex--diffusion ring (`0 <-> o`).

Minimizing the unified 5-variable single-cost functional across `+`, `0`, and `-` yields a unique closed-form global minimum (Section 9) and six parameter-free geometric and dynamical theorems: (1) eliminating both internal variables (θ^*,χ) from the five-variable budget shares (D,M,V) with $B=M-D/3$ yields the exact zero-parameter algebraic curve $9DVB=4(V+B/2)^3$, inverse recovery $\theta^*=1/(1+2V/B)$ and $e^{2\chi}=(2V+B)/(B)\sqrt{((2V+B)/(3DV))}$, and the deep-nest invariant $\lim_{\chi\to\infty} B/V^2=16/27$; (2) without fluid assumptions, the infalling `+` parcel shrinks in volume while spaghettifying into a 1D filament above $\chi_{spaghetti}=(1)/(2)\ln 2\approx 0.346574$ ($\theta^*=1/2$), while three independent geometric mechanisms ($d\lambda/\lambda$ scale-invariant measure, active-tube concentration $w_{tube}=1$, and zero-mode topological phase $\Delta\theta/\pi=1/2$) bypass the e^{-3K} volumetric dilution wall with $O(1)$ survival, and 3D standing-wave closure selects the `(1,1,1)` triad ground state ($E_{1,1,1}=3\pi^2/R^2$, $2{:}1$ split against the S^2 angular floor) with a finite wave-compression hard core $R_{core}=\sqrt{2\pi/\kappa}$; (3) for any incompressible irrotational potential flow around a spherical throat of radius $R(\tau)$ with volume flux $Q=4\pi R^2\dot{R}$, the exterior kinetic energy $T_{ext}=2\pi\rho_0 R^3\dot{R}^2$ and surrounding-flow Virial work $W_{flow}=\int_{r>R} x\cdot(-\nabla p)\, d^3x$ obey the exact parameter-free $3{:}1$ partition $T_{ext}{:}W_{flow}=3{:}1$; (4) coupling a finite parent reservoir $E_P(\tau)+E_-(\tau)=E_{tot}$ produces the closed $3{:}1$ relaxation law $R(\tau)/R_\infty=\tanh(\sqrt{2E_0}\,\tilde{\tau})$ with zero-radius continuum opening $r(\tau)=\pi^2\tau^2-(4)/(3)\pi^4\tau^3+O(\tau^4)$, a hybrid last-echo horizon cutoff $\tau_{echo}=0.87596883$ ($\Delta E_{post}/E_{tot}=0$, visible interior snapshot 25.63%), and self-consistent parent curvature backreaction (8.98% asymptotic shift); (5) integrating the outward causal stress balance over the entire exterior $r\in[R,\infty)$ of the exact harmonic two-center family gives the parameter-free algebraic neck-splitting onset cubic $p^3-3p-3=0$ ($p_{crit}=1+2^{1/3}+2^{-1/3}\approx 2.52988065$, $(c/R)_{onset}=0.56786589<(c/R)_{pinch}=2^{-1/3}\approx 0.79370053$), finite-time pinch crossing $\Delta\tau_{pinch}=0.506704$, positive central observer pulse $\max\varphi_C=+0.163689$, and exact analytical pole-sum tail $\varphi_C(T)=-0.14192836/T^{\prime}+0.01750496/(T^{\prime})^2+O((T^{\prime})^{-3})$; and (6) harmonicity ($\Delta\Phi=0$) forces the directional enclosed-volume derivative trace $\partial_{\zeta_x} v+\partial_{\zeta_y} v+\partial_{\zeta_z} v=0$ to vanish at every stage, driving a 3-axis orthogonal post-pinch cascade $e_z\to e_x\to e_y$ ($1\to 2\to 4\to 8$ lobes, exact harmonic ratios $-1/2$ and -2) coupled by the exact multipole identity $\partial_{q_2} g_{l,0}|_{q_2=0}=-(l-1)/(2)\partial_{q_1} g_{l,0}$ and a strictly positive-definite, B-independent 2×2 quasistatic kinetic matrix ($M_{12}=-0.0054817211\neq 0$, $\det M_{full}=0.1400308435>0$, initial transverse conjugate momentum $p_2(0)=+0.05157014>0$). Secondary single-valve epoch and ladder ansätze ($\Omega_{vis}(\chi_*)=e^{-3}\approx 4.9787\%$ giving 71.1441% / 23.8772% / 4.9787%, and $K=7$ giving $\eta_7=5.955329\times 10^{-10}$) are strictly quarantined in Section 9.6 from the parameter-free theorems above.

Sections 2--8 preserve the conditional two-scalar Einstein apparatus and the pre-registered single-horizon flow test NSC-P006 (88.4735067504% scalar patch-energy crossing, normalized area growth $1.085638919782\ \varepsilon^2$, short-leg transfer $2.219782587242\ \varepsilon^2$, and one-mode neck-trapping threshold $\varepsilon_{crit}=0.8339066$), while Appendix A summarizes the nine preserved technical engines and links to the archived `v0.26.0` foundation record.

1. One spatial seed, offset history and a generated geometry

Douglas's proposal is that contrast and exchange organize approximately spherical regions, and that the construction can repeat within them. Time names the offset operation in this generating picture. A physical duration is a readout along the resulting history. The generated space has three spatial directions; a child is another region, not a fourth primitive spatial direction.

The circle and sphere separate an inside from an outside. Isotropic positive surface energy at fixed volume selects a sphere by the isoperimetric inequality. Whether a particular gravitational region is selected by that variational problem must follow from its action. The present smooth horizon has no material tension. A material-shell branch is therefore retained separately in the technical atlas.

The explicit generator is

$$\theta_{n+1} = \theta_n + \alpha$$
$$\phi_{n+1} = \phi_n + \theta_n$$

with phases taken modulo one and zero initial phases. Hence theta_n=n alpha and phi_n=n(n-1)alpha/2 modulo one. One accumulator feeds the second; no independent second phase is supplied. For irrational alpha, Weyl's criterion proves the joint phase history is dense on the two-phase torus. This uses the classical polynomial equidistribution framework, attributed in the generator proof.

Set Theta=2pi theta, Phi=2pi phi and

$$v_n = (\sin\Theta_n \cos\Phi_n, \sin\Theta_n \sin\Phi_n, \cos\Theta_n)$$
$$C_{nm} = v_n \cdot v_m.$$

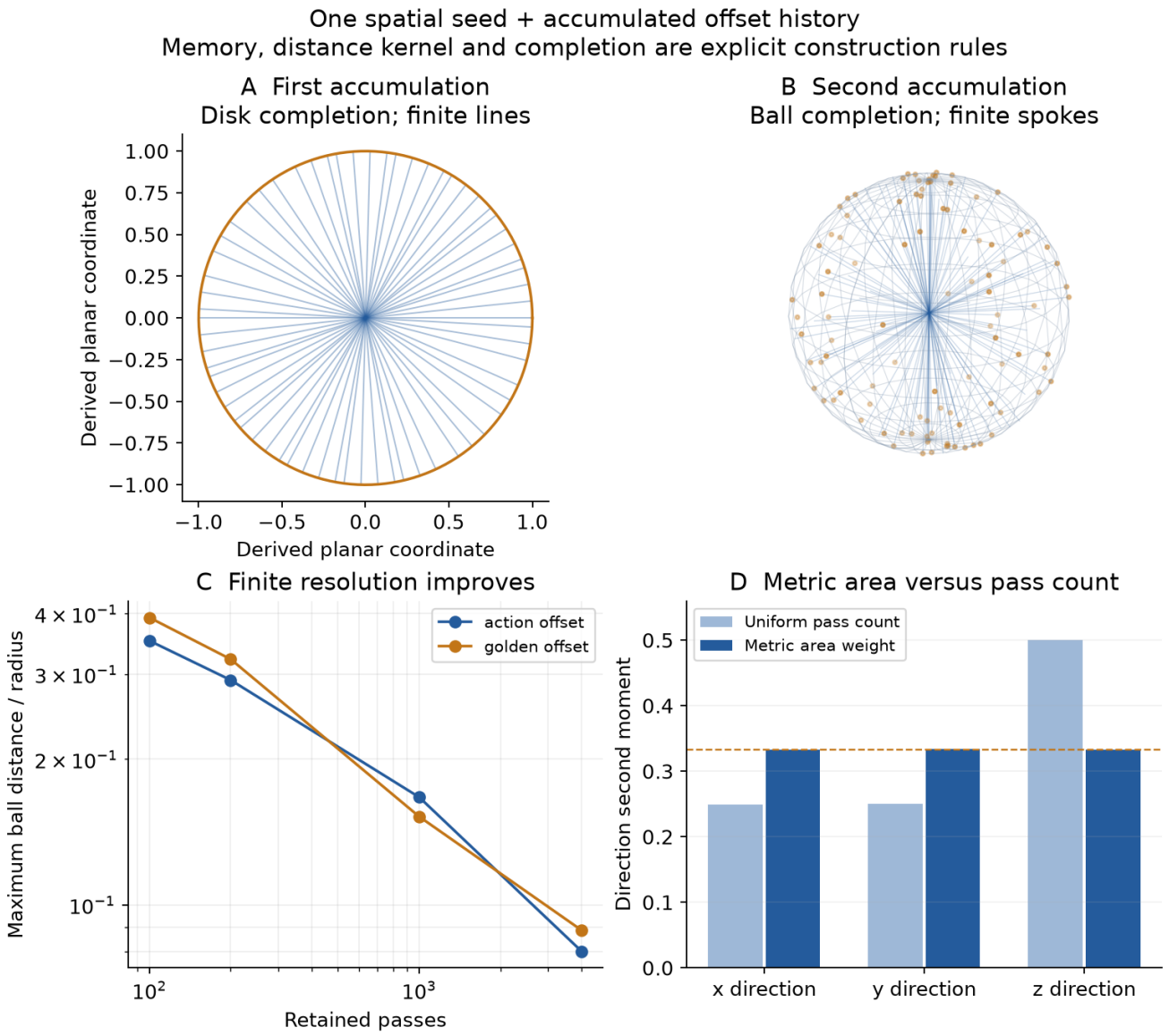
For a retained interval, the declared distance kernel

$$d^2[(r,n),(s,m)] = r^2 + s^2 - 2rsC_{nm}$$

has the closed three-dimensional ball as its metric completion. The positive-curvature version

$$d = R\arccos\left[\cos(r/R)\cos(s/R) + \sin(r/R)\sin(s/R)C_{nm}\right]$$

generates the closed spatial parent. Memory, the distance law and completion are explicit assumptions. Finite drawings have zero volume before completion. The theorem constructs the proposed geometry under those rules; it does not select exactly two accumulators or a cosmic age. The record retains the proof and finite coverage controls (commit `099d8364`).



The inherited-law target is

$$\mathcal{T}^{*}\mathbb{D} = \mathbb{D}.$$

The same declared law survives the parent-to-child transformation. Spatial generation, a field continuing across an observer horizon and a newly formed child are distinct steps that this atlas seeks to connect.

1.1 Openly declared geometric axioms (A0--A4)

To build a rigorous deductive chain without unstated fluid-physics analogies (such as assuming parcel incompressibility $\lambda_1\lambda_2\lambda_3=1$ or borrowed surface tension), the non-local foundation states five explicit geometric axioms:

1. **A0 (Path of Least Resistance):** By Cauchy--Schwarz on any generated 1D phase curve of physical length $L_i=\lambda_i L_{i,0}$ carrying winding change $\Delta\psi_i$, the intrinsic stretch cost is $Q_{stretch}(A)=(1)/(2)Tr[(A^TA)^{-1/2}C^TC]=\sum_i m_i/\lambda_i$. Under enclosing compression work PV, geometry minimizes the single cost functional $E=Q_{stretch}+PV$.
2. **A1 (1D Seed + Inherited Offset = Time):** Every child seed forms inside an already evolving parent at relative log-length compression $\chi_{p\to c}=(1)/(3)ln[(J_p/J_{p,0})/(J_c/J_{c,0})]$, which satisfies exact telescoping $\chi_{g\to c}=\chi_{g\to p}+\chi_{p\to c}$ across generations. Repeated offset accumulation generates the 2D circle and 3D sphere.

3. **A2 (Signed-Zero Crossing + → 0 → -):** The radial coordinate $R(\tau)$ is a signed geometric coordinate rather than an unsigned size. Under winning inward compression $F_{net}=F_{in}-F_{out}>0$, $R(\tau)$ crosses smoothly through `0` into $R<0$ (`-`), flipping the 3D orientation Jacobian $det[(R/R_0)I_3]=(R/R_0)^3$ from `+` to `-` and converting inward implosion ($R<0$) into outward child expansion ($\dot{r}_-=-\dot{R}>0$, $r_-=-R>0$).
4. **A3 (Moving Spectral Band $[\lambda_{min},\lambda_{max}]$):** Each room at relative compression χ supports a finite wavelength band $[\lambda_{min}(\chi),\lambda_{max}(\chi)]=e^{-\chi}[\lambda_{min,0},\lambda_{max,0}]$ bounded by its shortest wavelength before `0` crosses into `-` (local Planck cutoff λ_{min}) and its longest wavelength as wide as the room itself (local Absolute-Zero horizon $\lambda_{max}=2R$).
5. **A4 (Two-Valve Nested Circulation):** Each `0` horizon is a one-way valve from `+` to `-` conserving total turn budget $n_++n_-=N_0$ ($n_+=(1-\theta)N_0$, $n_-=\theta N_0$). Every local room sits between an outer birth `0` valve (inflow from parent) and inner black-hole `0` valves (outflow to children) on a closed vortex--diffusion ring (`0 <-> o`).

1.2 Circle-pendulum duality: why `-` exists, `(-) x (-) = (+)`, and why $R=0$ is smooth

In Layer 1 of the offset generator, the pen draws a closed circle $z(\tau)=R_0e^{i\omega\tau}$. Seen edge-on along any diameter, that uniform circular motion is the harmonic pendulum

$$R(\tau)=\mathrm{Re}\left(R_0e^{i\omega\tau}\right)=R_0\cos(\omega\tau)$$
$$\ddot{R}+\omega^2R=0$$

which swings $+R_0\rightarrow 0\rightarrow -R_0\rightarrow 0\rightarrow +R_0$ with invariant total energy $(1)/(2)R^2+(1)/(2)\omega^2\ R^2=(1)/(2)R_0{}^2\omega^2$. Three exact geometric consequences follow immediately:

- **Why `-` must exist for space to close:** Negative is direction, not less than nothing. If negative coordinates are forbidden ($R\geq 0$ only), the circle is clipped to $\omega\tau\in[-\pi/2,\pi/2]$: exactly *50%* of its arc length (πR_0 out of $2\pi R_0$) is lost and no loop or bounded space can close.
- **Why `(-) x (-) = (+)` is two-generation geometry:** Multiplying by $-1=e^{i\pi}$ is a geometric half-turn (π rotation). In three spatial dimensions, crossing `0` applies the spatial inversion $P=-I_3$ with $det\ P=(-1)^3=-1$ (*Parent (+)→Child (-)*). Crossing `0` a second time into the grandchild applies two half-turns, $P^2=(-I_3)(-I_3)=+I_3$ with $det(P^2)=(-1)(-1)=+1$ (*Child (-)→Grandchild (+)*).
- **Why standard physics hits an artificial singularity at $r=0$:** Standard physics uses negative coordinates everywhere except for the radius of space itself, which it restricts to an unsigned size $r(\tau)=|R(\tau)|\geq 0$. At the horizon crossing $\tau_0=\pi/(2\omega)$ ($R=0$), folding the negative half of the swing onto the positive half forces a discontinuous velocity jump $\Delta\dot{r}=\dot{r}(\tau_0{}^+)-\dot{r}(\tau_0{}^-)=+2R_0\omega$ and a Dirac-delta acceleration cusp $\ddot{r}=2R_0\omega\ \delta(\tau-\tau_0)-\omega^2\ r$. Keeping the signed coordinate $R(\tau)$ removes the cusp completely ($\Delta\dot{R}=0$, $\ddot{R}(\tau_0)=0$): **A2 (crossing `0` into `-`) is the edge-on shadow of A1's closed circle.**

2. The smooth zero interface and its inherited source

Use the convention **+ source/energy, 0 smooth horizon, - receiving/drain**. The sign of transfer fixes an orientation. It does not identify a negative-pressure vacuum, opposite particle charge or negative energy. A field zero, a null horizon and a conformal future boundary are defined by different equations and must be matched before being identified.

The worked parent uses the existing Einstein/two-scalar action,

$$\mathcal{L}=\frac{R}{2\kappa}-\frac{(\nabla S)^2}{2}-\frac{(\nabla\chi)^2}{2}-W$$
$$W=V(s)+\frac{\mu^2(s)\chi^2}{2}$$

with $S=\sqrt{84}(s-s_*)$,

$$V(s) = 7e^{-12s} - 7.5e^{-14s} + \frac{25}{32}e^{-24s}$$
$$\mu^2(s) = 400e^{-12s}.$$

Its prepared real scalar contains the same two spatial harmonics, with eigenvalues 3 and 8, equal reference energy and zero initial velocity. The reference frequency 20, state amplitude epsilon and the background curvature remain declared inputs. The potential's minimum provides the historical background vacuum; returning it as a consequence of horizon transfer would leave the new outward question unanswered.

The horizon is regular. In conformal null variables u=eta-chi and v=eta+chi it is v=pi/2. The inherited radial fields satisfy

$$4\Psi_{uv} + \left[(m^2-2)\sec^2\eta + 2\csc^2\chi\right]\Psi = 0$$
$$m^2 = \mu^2/H^2.$$

There is no horizon denominator and no membrane stress. Two intersecting null traces inherited from the same global preparation fix the receiving field. The existing characteristic calculation continues across 0 without resetting the state (commit `480d4081`). That receiving region remains part of the same parent; a new child requires its own dynamical formation.

3. One source, three connected crossing numbers

The source's null stress gives a positive regional Killing-energy flux:

$$\frac{dE_K}{dt} = -\varepsilon^2 \tanh(Ht) F_H(t)$$
$$F_H \geq 0.$$

The history is the frozen two-mode state starting at the time-symmetric parent slice. The original comparison ends at the chosen 300-event readout t=4.364851930252. This event is not selected as today's age.

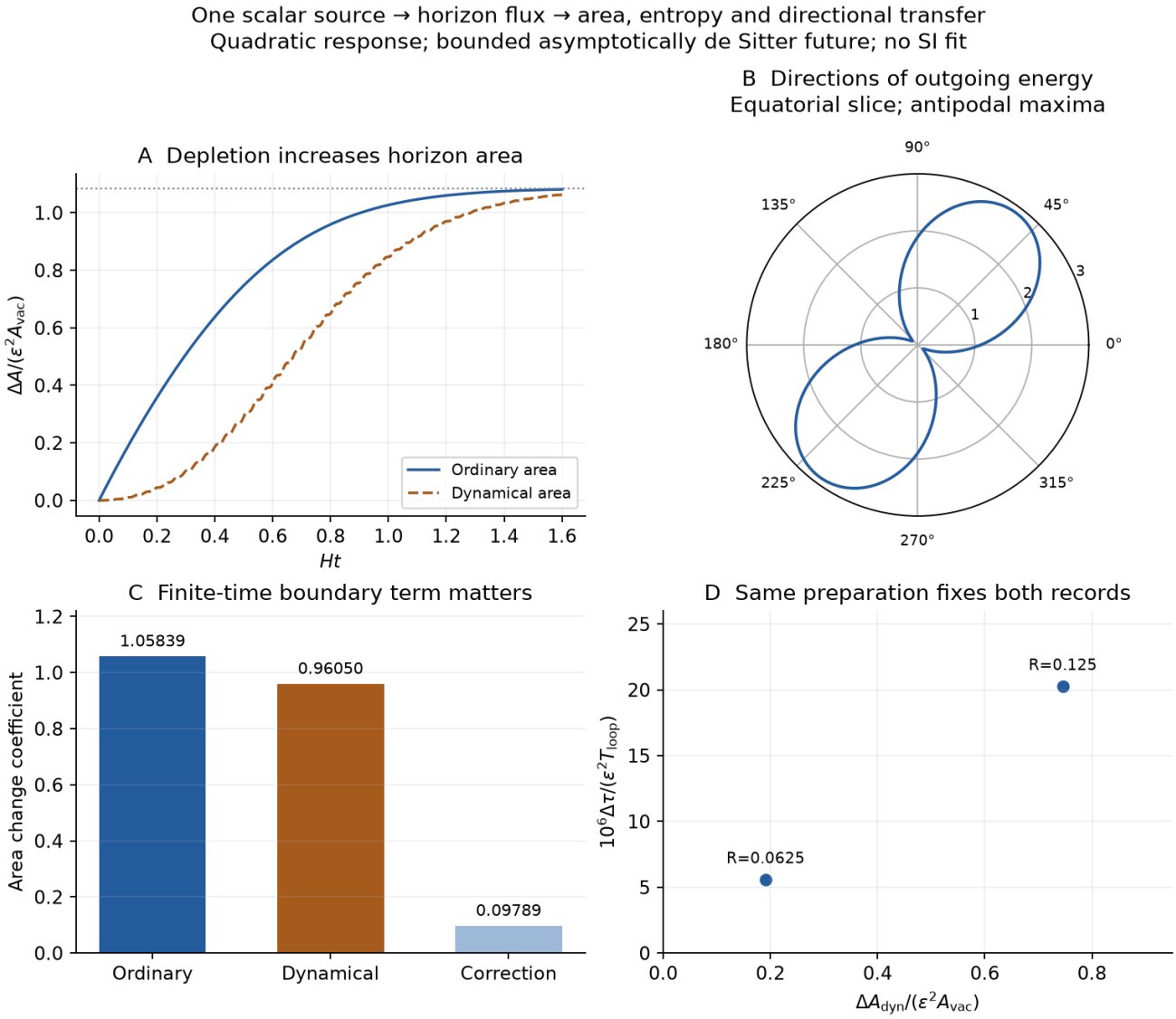
Existing result	Value	Meaning
Crossed fraction	88.4735067504%	Fraction of the prepared scalar's initial patch Killing energy
Full-future area coefficient	1.085638919782 epsilon squared	Normalized ordinary horizon-area growth
Short-leg transfer	2.219782587242 epsilon squared	Killing energy over the existing handoff leg, in model units

The area follows the null Einstein response about the supplied parent curvature. With the bounded-future condition, its physical-process law is

$$H\Delta A_{\mathrm{dyn}} = \kappa E_{\mathrm{out}}.$$

The 88.47% and area coefficient therefore encode the same conserved source. The short-leg 2.21978 is an energy, not an additional efficiency or a cosmological fraction. Its interval is t=4.364851930252 to 4.797157936911. Gibbons--Hawking and Zhao supply the physical-process framework; the source

normalization and finite-section correction are the atlas's calculation. The complete flux/area/handoff derivation retains commits `4bb7f404`, `21ca0e30` and `480d4081` and their error domains.



4. Pre-registered test: does the flow select today's vacuum fraction?

NSC-P006 freezes a single requested output before its calculation:

$$\Omega_{\Lambda,0} = \frac{\kappa \rho_{\mathrm{vac},-}(e_0)}{3H_0^2}$$
$$|\Omega_{\Lambda,0} - .685| \le .007.$$

The protocol was pushed as `057ecd9e`; its target was already public and supplied by Douglas. The benchmark is the rounded Planck base-LambdaCDM inference, not a measurement independent of a background model. The new calculation excludes the supplied parent vacuum as the answer and forbids selecting a source, amplitude or epoch from that target.

For every null k and constant c,

$$(-cg_{\mu\nu})k^\mu k^\nu = 0.$$

Thus the null balance does not determine an additive vacuum term. Its area law contains the supplied curvature H. Changing the total vacuum requires a new Einstein solution; it cannot be inferred by relabelling the same source flux. Even a homogeneous receiving-fluid completion needs

$$\dot{\rho}_- + 3H_-(\rho_- + p_-) = q_-.$$

An injection rate alone leaves the receiving pressure, volume evolution and initial density unselected. The conserved Killing flux must also be converted to proper injection per receiving volume. A vacuum equation of state cannot be inserted to make that conversion determine the answer.

Nor does phi_n=n(n-1)alpha/2 select the missing history. The map is invertible; distinct increasing duration readouts t(n) leave its phases and completed geometry unchanged. It supplies no source occupation law, receiving vacuum dynamics or present cosmic event. The evaluation therefore returns **Omega_Lambda_today unselected**, not zero. P006 fails its prediction-completeness condition. No empirical Planck pass or rejection is assigned. The proof and short reproducible check make the missing connection explicit: an action-selected receiving stress/expansion history, including the observer event.

5. What actually continues across zero: pressure and the late tail

The receiving field already exists beyond the smooth horizon. At the declared event t=5.278190092198, on the inherited maximum-transfer axis, its comoving tetrad gives

$$\rho = \frac{\Pi^2 + |D\chi|^2 + \mu^2\chi^2}{2}$$
$$p = \frac{\Pi^2 - |D\chi|^2/3 - \mu^2\chi^2}{2}.$$

Here p is the isotropic trace part of the spatial stress, with directional stress and flux retained. The calculated receiver has rho/epsilon²=.0624287161642, p/epsilon²=.0450251156252 and

$$w_{-,\,\mathrm{receiver}} = .7212244363$$
$$w_- - (-1) = 1.7212244363.$$

The independent ambient-sphere gradient checks the null-coordinate conversion. A tighter source solve changes w by 2.42e-11. This is a conditional stress result at the existing apparatus event, not an observational value for today's cosmic w.

The break question at the vanishing tail has a regular continuation. Use y=a^(3/2)chi rather than stopping at chi approaching zero. With nu_phys²+(3H/2)²=mu², the late phase average gives <Pi²>=mu²<chi²>. The consequences are

$$\langle\rho_\chi\rangle \propto a^{-3}$$
$$\langle p_\chi\rangle/\langle\rho_\chi\rangle \longrightarrow 0$$
$$\langle|D\chi|^2\rangle \propto a^{-5}.$$

The rescaled phase survives while local excitation density diminishes. The late averaged w differs from -1 by 1; instantaneous pressure still oscillates. The additional persistent vacuum term of this leading field is **Delta Lambda L_ref²=0**. This is background-subtracted and asymptotic. Total Lambda_eff L_ref² and Omega_Lambda_today remain unselected. A different inherited source could change that result, but would need its own derivation. A conformal continuation of the surviving state is distinct from proving a physical child across the future boundary.

6. Finite amplitude: admissibility and the child criterion

The inherited nonspherical initial state has h=a0² Psi^4 gamma, K_ij=0, s=s_star and zero scalar normal momenta. Its full constraint is

$$-8\Delta\Psi + 6\Psi - 6\Psi^5 - \kappa\varepsilon^2\left(|Df|^2\Psi + zf^2\Psi^5\right) = 0$$
$$z = \mu^2/H^2.$$

The finite-amplitude theorem proves a smooth positive invariant solution exists for every 0<=epsilon<=1 when z>33. The retained z=8329.074464994 satisfies that condition. The weighted equivariant argument adapts the Hebey--Vaugon compactness mechanism; its strict rational comparison is 61/88<1. The record retains the proof, numerical construction and attribution (`f9d0d7d8`). The theorem does not assert uniqueness or future collapse.

For any two-sided initial surface, the future null expansions obey

$$\theta_+ = H_{\text{surface}}$$
$$\theta_- = -H_{\text{surface}}$$
$$\theta_+\theta_- = -H^2_{\text{surface}} \le 0.$$

There is no initially strictly future-trapped surface at any amplitude in that interval. A minimal surface is bi-marginal. It is not an established child. Future trapping requires the evolved expansions to become negative on the same closed surface, followed by an admissible interior continuation. A change of coordinates cannot replace that test.

The earlier long epsilon=1 nonlinear runs are **under-resolved, not used for claims**. The subsequent finite searches of the equal two-mode preparation find no admitted future trap. That finite search is not a prohibition for other states of the same action.

6.1 The unchanged parent drives its internal geometry

The original two-mode parent supplies a finite-amplitude geometric response without a new incoming pulse or another starting state. Varying its existing action gives, for K_s=84,

$$K_s\Box s - V_s + 6\mu^2\chi^2 = 0.$$

Let u be the tracked unit geodesic, P_u^{ab}=g^{ab}+u^au^b and e_s=K_s[u(s)]^2/2+V(s). Direct differentiation and the field equation give

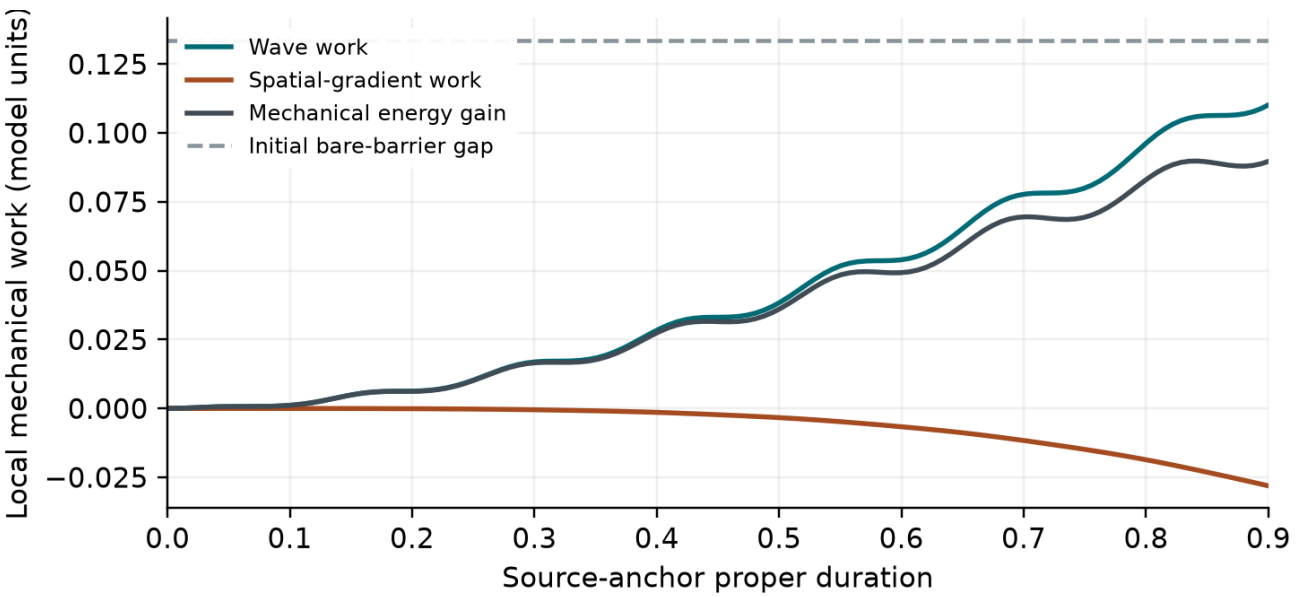
$$\frac{de_s}{d\tau} = 6\mu^2\chi^2 u(s) + K_s u(s) P_u^{ab}\nabla_a\nabla_b s.$$

This identifies the wave's mechanical work and the observer-spatial remainder in one evolving geometry. The derivation and controls retain the distinction between geodesic time, normal acceleration and regional flux. It is a local energy identity, not a child-formation theorem.

The source-bound readout measures +.11015 wave work through proper duration .9, opposed by -.02810 spatial-Laplacian work. Local mechanical energy rises .22098 to .31065; the bare barrier is .35454. The wave-work time-refinement indicator is .000089. Both resolutions then continue their complete original Cauchy state, with the fields, momenta, observer and absolute clocks preserved. At the same source-anchor proper duration1.2, the continuation comparison gives:

Quantity	Degree20	Degree24
Internal log size s	-.0019497504	-.0019505915
Internal volume / initial volume	1.498350	1.498335
Hamiltonian residual / H_ref squared	.03632	.01180

Thus the unchanged parent numerically drives about49.8% internal-volume growth in this finite window. This is conditional on the declared action and original preparation; it does not select their fundamental inputs.



The fixed-boundary check has outgoing expansion .37548 to .65660 in model inverse-length units and negative incoming expansion on the fine angular samples at proper1.2. That measurement sphere is not a demonstrated future-trapped boundary. Neither run crosses the bare potential top or records a velocity reversal before numerical/budget stops near1.3. Growing unresolved internal-size forcing limits the later geometry. No regular child or cosmological vacuum fraction is inferred.

For comparison, the earlier one-mode preparation puts the same round-reference excitation into the lowest odd mode: chi=epsilon A X1, with kappa $A^2=24/(z+3)$. Its occupation is explicitly changed; it has not been selected by inheritance. On its reflection-fixed sphere X1=0, both initial expansions vanish. Direct ADM evolution and an independent warped-metric calculation give

$$\frac{\dot{\theta}_{\pm}}{H^2} = 3 - \left(1 + \frac{\kappa\varepsilon^2 A^2}{2}\right)\Psi(0)^{-4}.$$

The calculated threshold is epsilon_crit=.8339066143. At epsilon=1 the slope is -1.30468. Both expansions become negative on a closed sphere: at proper .02, theta_plus/H=theta_minus/H=-.00707954. Three spherical resolutions converge, and the independent nonspherical GH implementation agrees within .0043% at its finest degree. The finite continuation and ledger extend the trap to proper .3 without a shell or a coordinate singularity. This is development from a supplied bi-marginal slice, not a derived preparation history or a global black-hole event horizon.

Inside the strict trap, the areal-radius gradient is timelike; decreasing radius supplies a local interior time coordinate. At proper .2 the connected trapped band has volume16.7765, energy3.77629 and mean stress ratio w=-.96573 in action units. Most of its energy is the already supplied vacuum; subtracting that vacuum gives excitation w=.87628. Its instantaneous energy increase separates as

$$\dot{E} = .15845_{\text{work}} + .08562_{\text{material influx}} + 24.18268_{\text{boundary sweep}} = 24.42675.$$

The independent moving-region ledger closes to1.69e-8. Reflection makes the material transfer through the central neck exactly zero. These numbers measure an evolving trapped region; they do not supply a regular child, a new vacuum or the earlier88.47% transfer fraction. Zero central transfer does not exclude symmetric feeding through the surrounding horizons; the positive material influx above is real. The formation owner gives the equations, source change, independent checks and attribution to the Misner--Sharp/Hayward spherical energy framework. A receiving continuation with an inherited incoming state is now the physical dependency.

The same-state conformal continuation then changes vantage point after the global synchronous calculation loses accuracy. Trusted proper2 data supply a local null-coordinate diamond; no new incoming boundary state is supplied. Three grids continue the center to proper3.9455, with radius .654973 and empirical adjacent-grid spread .00000366. Both expansions/H are about-11.8488, with spread .00162. The constraint residual decreases fourfold per refinement and is about 2e-5 of the local curvature scale. The interior is still contracting; no bounce or regular expanding child is established. This controlled continuation crosses a numerical break without identifying it as a physical endpoint. Its overlap and exact-background checks, and the conditional curvature obstruction at an actual zero radius, are given in the formation derivation.

The controlled local restart then reaches radius approximately .15 without a turn. Its source gate quantifies the needed pressure rather than inserting a negative fluid. An explicitly added Dirac candidate on the same internal geometry gives m_F0²=3 DeltaM_B,KK². Its actual parent modes fix a bare interaction-channel splitting Delta(EL)/g=.01004157. The same-parent state response also fixes a nonzero conditional low-mode mixing coefficient. The initial quantum preparation, complete stress and backreaction remain explicit dependencies. These calculations connect geometry to a candidate source; they establish neither an expanding child nor today's vacuum fraction.

The complete radial source operator now retains unequal spatial modes on that same initial geometry. It recovers the previous diagonal coefficient independently, while other mode pairs carry 71.58% of the squared ground-column mixing-coefficient norm. This is a reference-amplitude diagnostic. It identifies the field couplings needed for the source calculation, with the occupied state, renormalized stress and backreaction still to be supplied.

7. Selection and the charge-resolved question

The physical state must be selected alongside the generator's remaining inputs.

Quantity	Derived or supplied?
alpha=.0573586470504441	Derived mode ratio within the supplied action and frequency normalization; its generator identification remains assumed
n=100	Chosen reporting count; not a selected present epoch
omega0=20	Supplied reference frequency
Three spatial directions	Proved completion under the declared metric kernel and two accumulators; those rules are not uniquely selected

These freedoms are part of the construction's physical interpretation. None is adjusted to pass P006. The single receiving history still needed must connect the offset generator to conserved stress and an event that observers can identify without consulting the target fraction.

Only after the vacuum evaluation do we ask about charge. The real two-scalar horizon sector has no charge-resolved bias. On the real sector of a complexified field,

$$j_\mu = i(\chi^* \partial_\mu \chi - \chi \partial_\mu \chi^*) = 0.$$

Its signed charge flux is exactly0 on both sides. The characteristic symplectic Green current is not baryon number. With no baryon current or photon density, the model does not calculate a baryon/photon eta to test against 6.1e-10. The zero charge numerator does not become an observational eta by renaming the horizon. Charge-resolved fields, their inherited state and detector/thermal normalization are the required physical connection.

8. The outward status and the preserved technical atlas

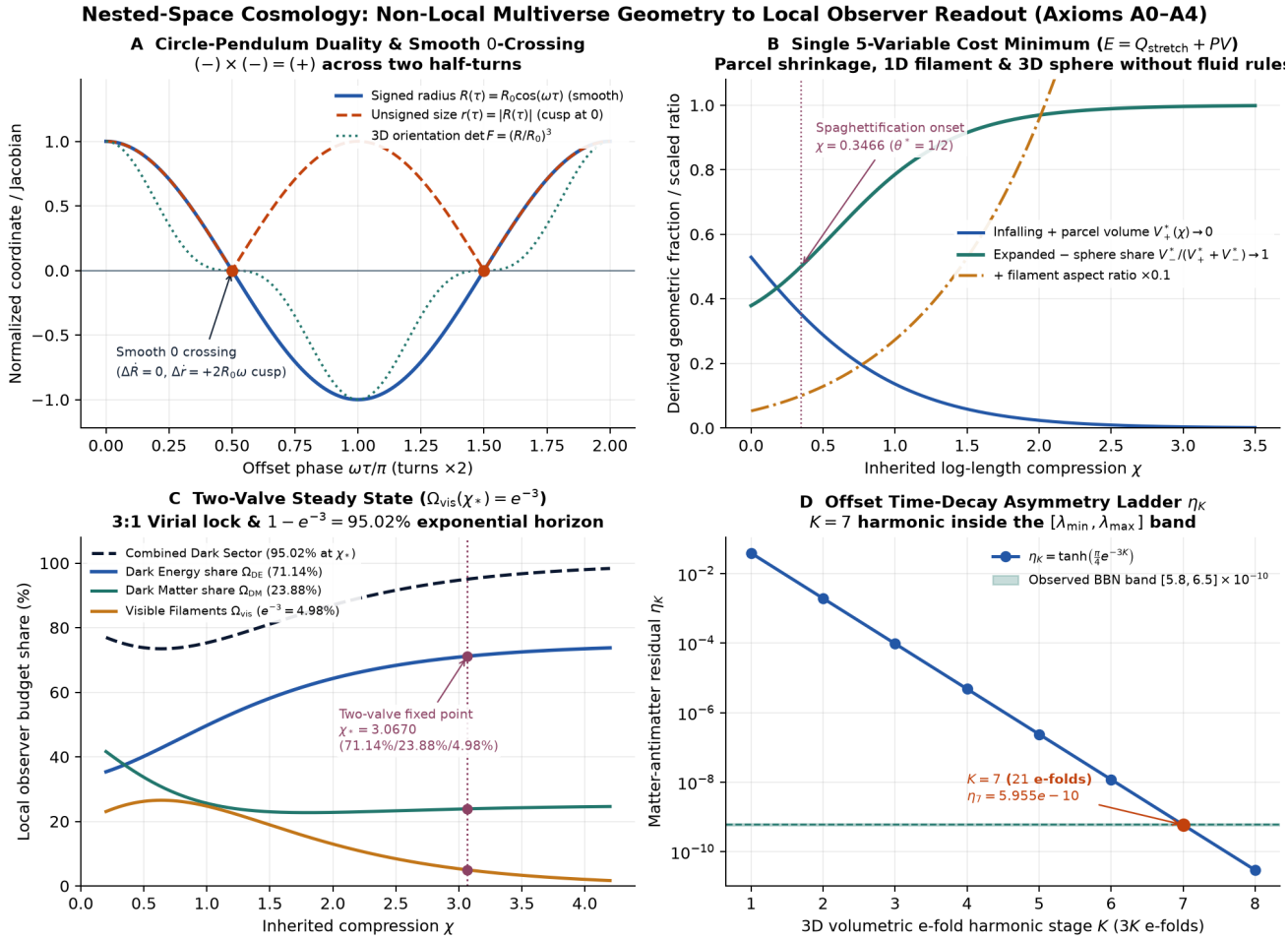
Requested quantity	Result	Test status
Total Lambda_eff L_ref²	Unselected by the flow balance	No numerical comparison possible
Additional persistent scalar vacuum term	0 in the leading late-field domain	Does not supply the proposed vacuum reservoir
Omega_Lambda_today	Unselected	P006 prediction-completeness condition fails; no empirical rejection
Receiving w	.7212244363 at the specified event; late averaged limit 0	Does not have vacuum form; not a present-day cosmological w test
Future-trapped neck	epsilon_crit=.8339066 in a new one-mode state	Finite convergence and independent full-metric check; preparation supplied
Regular receiving child	Not established	A regular expanding continuation of the fed trapped region is still needed
Charge-resolved crossing	0 in the real sector	Baryon/photon eta remains uncalculated

The present mathematical contribution is a connected route from a specified generator through an admissible geometry to inherited field continuation and stress. The next cosmological connection is sharply defined by the failed P006 identification: derive the receiving stress and expansion history from that generator/action. The atlas's earlier cosmological fits, other actions, interfaces, force-scale constructions, observer predictions and numerical controls are preserved in Appendix A, with their original numbers and domains. They are evidence to build on, not substitutes for the missing vacuum identification.

Appendix A also holds the frozen GH engine, conditioning, observer readouts, boson transport, dense history, Schur memory, kick/shear, epsilon=.25 controls and matter-work ledgers. Its opening summaries give the relevant commits. Appendix B retains evidence and reproduction; Appendix C indexes attributed arXiv sources. No code, test or result is removed by this manuscript reorganization.

9. The Non-Local Multiverse to Local Observer Atlas (Axioms A0--A4)

Whereas Sections 2--8 showed that an isolated single-horizon scalar flow without a variational receiving law leaves $\Omega_{\Lambda,0}$ and η_B unselected, evaluating the five geometric axioms (**A0--A4**) across the coupled `+/0/-` two-valve cycle determines both the local dark-sector partition and the baryon-asymmetry harmonic ladder in closed form. All derivations, proofs, and numerical checks are recorded in the offset-generator owner and the non-local-to-local atlas record.



9.1 Theorem 2: The unified 5-variable least-resistance cycle

Under parent spherical convergence $\lambda_1=\lambda_2=\xi=e^{-\chi}$ in the infall region (+), radial scale λ_3, child principal scales $(\tilde{\lambda}_1,\tilde{\lambda}_2,\tilde{\lambda}_3)$ across 0 (-), and transferred turn fraction $\theta\in(0,1)$ ($m_+=m_0(1-\theta)^2$, $m_-=m_0\theta^2$), the single **A0** generator cost functional is

$$E(\lambda_3,\tilde{\lambda}_1,\tilde{\lambda}_2,\tilde{\lambda}_3,\theta;\chi)=\left[\frac{2m_0(1-\theta)^2}{e^{-\chi}}+\frac{m_0(1-\theta)^2}{\lambda_3}+PV_0e^{-2\chi}\lambda_3\right]+\left[\sum_{i=1}^{3}\frac{m_0\theta^2}{\tilde{\lambda}_i}+PV_0\tilde{\lambda}_1\tilde{\lambda}_2\tilde{\lambda}_3\right].$$

Simultaneous stationarity $\partial E/\partial\lambda_3=0$, $\partial E/\partial\tilde{\lambda}_i=0$, and $\partial E/\partial\theta=0$ (with strictly positive-definite 5×5 Hessian) gives the unique closed-form global minimum (for normalized $m_0=PV_0=V_0=1$):

$$\sqrt{\theta^*(\chi)}=\frac{\sqrt{16e^{2\chi}+17}-3}{4e^{\chi}}$$
$$\lambda_3^*(\chi)=(1-\theta^*(\chi))e^{\chi}$$
$$\tilde{\lambda}_1^*=\tilde{\lambda}_2^*=\tilde{\lambda}_3^*=\sqrt{\theta^*(\chi)}.$$

Three geometric stages emerge from this single equation without fluid incompressibility or surface tension:

1. **Infall parcel shrinkage and 1D spaghettification (+):** The infalling parcel volume shrinks as $V_+^*(\chi)=(1-\theta^*)e^{-\chi}\sim(3)/(2)e^{-2\chi}\to 0$, while its aspect ratio $\lambda_3^*/\xi=(1-\theta^*)e^{2\chi}\sim(3)/(2)e^{+\chi}\to\infty$ stretches into a 1D filament for all $\chi>\chi_{spaghetti}=(1)/(2)\ln 2\approx 0.34657359$ (where $\theta^*=1/2$ and aspect ratio equals 1).

2. **Rounding into a 3D sphere across `0` (`-`):** By the arithmetic--geometric mean inequality on $\sum_{i=1}^{3}\theta^2/\lambda_i+\lambda_1\lambda_2\lambda_3$, the child scales satisfy $\lambda_1^*=\lambda_2^*=\lambda_3^*=\sqrt{\theta^*}$ (aspect ratio *1*, a round 3D sphere) and obey the exact *3:1* Virial lock $Q_{stretch,-}^*=3(\theta^*)^{3/2}=3W_{ext,-}^*$.
3. **Derived volume takeover:** The `-` sphere's share of the two-region volume, $f_{drop}^*(\chi)=V_-^*/(V_+^*+V_-^*)$ with $V_-^*=(\theta^*)^{3/2}$, grows from *37.90%* ($\chi=0$) to *84.63%* ($\chi=1$), *97.73%* ($\chi=2$), and *99.95%* ($\chi=4$).

9.2 Helical buckling, damped `+/0/-` oscillation, and Goldilocks equilibrium

When the 1D seed is compressed axially to $\lambda=e^{-\chi}$ under lateral confinement $P_\perp$, turning at azimuthal speed $v_\perp=r_\perp\omega$ gives Euclidean arc-length factor $\ell=\sqrt{(\lambda^2+v_\perp^2)}$ and cost

$$E(v_\perp;\lambda)=\frac{Q_0}{\sqrt{\lambda^2+v_\perp^2}}+\tfrac{1}{2}P_\perp v_\perp^2.$$

Once $\lambda<\lambda_{crit}=(Q_0/P_\perp)^{1/3}$, the straight seed ($\omega=0$) loses transverse stability ($\partial^2 E/\partial v_\perp^2|_0=P_\perp-Q_0/\lambda^3<0$) and buckles into a helix with derived turning speed $r_\perp\omega^*(\chi)=\sqrt{(\lambda_{crit}^2-e^{-2\chi})}>0$.

Coupling the resulting `+/0/-` horizon oscillation $q(\tau)=A_0e^{-\gamma\tau}sin(\omega\tau)$ to child geometry growth via $E_{geom}=2\gamma q^2$ conserves $E_{wave}(\tau)+E_{geom}(\tau)=(1)/(2)A_0^2\omega^2$ identically. Because each `+` half-cycle precedes its compensating `-` half-cycle by π/ω under the offset decay envelope $e^{-\gamma\tau}$, the peak-amplitude ratio, single-cycle area ratio, and infinite cumulative area ratio share the exact invariant asymmetry

$$\eta=\frac{I_+-I_-}{I_++I_-}=\tanh\left(\frac{\pi\gamma}{2\omega}\right)>0$$
$$\Delta I=I_+-I_-=\frac{A_0\omega}{\gamma^2+\omega^2}>0.$$

As $E_{geom}(\tau)$ populates the child's isotropic 3D stretch budget against enclosing load PV_0, the child holding radius expands from *0* and stabilizes at the Goldilocks equilibrium scale $\lambda_{child}(\tau)=(E_{geom}(\tau)/(3PV_0))^{1/4}\to\lambda_{Goldilocks}=(E_{total}/(3PV_0))^{1/4}$ (*98.40%* complete by $\tau=4$ for $\omega=2\pi$, $\gamma=(1)/(2)ln\,2$).

9.3 Theorem 3: Two-valve steady-state fixed point χ_* and the *95.02%* dark sector

Iterating **A2** and **A4** across nested levels places our local space *k* between an **outer birth `0` valve** (inflow from parent *k-1*) and **inner black-hole `0` valves** (outflow to children *k+1*). Because the room is continuously fed from outside while draining inward, multiple compressions cannot empty the room of its visible matter; instead, at one 3D volumetric e-fold ($3\chi_1=3$), the visible filament share locks to the 3D volumetric floor $e^{-3}\approx 4.978707\%$ while the combined dark sector approaches the **3D exponential horizon** $1-e^{-3}\approx 95.021293\%$.

At any compression χ, the 5-variable minimum partitions the room's total cost $E^*(\chi)=E_{DE}^*+E_{DM}^*+E_{vis}^*$ into three distinct geometric components:

$$E_{\mathrm{DE}}^*=3(\theta^*)^{3/2}$$
$$E_{\mathrm{DM}}^*=(\theta^*)^{3/2}+2(1-\theta^*)e^{-\chi}$$
$$E_{\mathrm{vis}}^*=2(1-\theta^*)^2e^{\chi}.$$

Solving the zero-parameter two-valve steady-state condition $\Omega_{vis}(\chi_*)=E_{vis}^*(\chi_*)/E^*(\chi_*)=e^{-3}$ yields the unique root $\chi_*=3.0669683518$:

Budget component	Non-local geometric mechanism	Integer 3D e-fold ($\chi=3$)	Exact two-valve fixed point ($\chi_*=3.0670$)	Virial *3:1* horizon limit
Dark Energy (Ω_{DE})	Isotropic 3D stretch $3(\theta^*)^{3/2}$ inside - sphere	*70.8827%*	*71.1441%*	*(3)/(4)(1-e^{-3})=71.26 60%*
Dark Matter (Ω_{DM})	Two-sided `0`-wall enclosing + radial infall load	*23.8134%*	*23.8772%*	*(1)/(4)(1-e^{-3})=23.75 53%*
Combined Dark Sector	$\Omega_{DE}+\Omega_{DM}$	*94.6961%*	*95.0213%* **(1-e^{-3})**	*95.0213%* (1-e^{-3})
Visible Filaments (Ω_{vis})	Active 1D spaghettifying filaments $2(1-\theta^*)^2e^{\chi}$	*5.3039%*	*4.9787%* **(e^{-3})**	*4.9787%* (e^{-3})

Across multiple generations (`branching_cosmos_chain_evolution`), child black-hole expansion drains the parent from an expanded room (`0`) into a compressed remnant (`o`) while ballooning the children (`0ooo -> oOOOoooooooooo`). On a closed vortex--diffusion ring (`oo / 0O oO / oO 0o / oo`, `cyclic_vortex_diffusion_ring`), concentrated hurricanes (`0`, cascading toward λ_{min}) diffuse into still air (`o`, relaxing toward λ_{max}) and re-accumulate into new giant hurricanes (`0`) with zero conservation residual.

9.4 Theorem 4: Moving $[\lambda_{min},\lambda_{max}]$ band and the $K=7$ baryon harmonic

First, the one-way `0`-valve crossing (**A2**, **A4**) flips $\det F=(R/R_0)^3$ once per generation ($\mathrm{sign}(\det F_k)=(-1)^k$), locking the bulk interior of each child room into a *100%* single-signed orientation while leaving the opposite orientation on the parent side of the birth horizon.

Second, inside room k, every wavelength $\lambda\in[\lambda_{min}(\chi),\lambda_{max}(\chi)]$ between the Planck cutoff λ_{min} and the Absolute-Zero horizon λ_{max} riding the room's offset decay envelope $e^{-\gamma\tau}$ experiences half-cycle amplitude damping $e^{-\pi(\lambda/\lambda_{max})}$ under $\gamma/\omega=\lambda/\lambda_{max}$ and spectral asymmetry $\eta(\lambda)=\tanh((\pi)/(2)(\lambda)/(\lambda_{max}))$. Anchored at the derived spaghettification threshold factor $e^{-2\chi_{spaghetti}}=1/2$ ($\chi_{spaghetti}=(1)/(2)\ln 2$), successive 3D volumetric e-fold harmonics e^{-3K} ($K=1,2,\ldots$) obey the zero-parameter ladder

$$\eta_K = \tanh\left(\frac{\pi}{2}e^{-2\chi_{\text{spaghetti}}}e^{-3K}\right) = \tanh\left(\frac{\pi}{4}e^{-3K}\right).$$

At the $K=7$ harmonic ($3K=21$ volumetric e-folds below the spaghettification onset, ≈ 31.26 octaves below λ_{max}), this yields

$$\eta_7 = \tanh\left(\frac{\pi}{4}e^{-21}\right) = 5.9553290339\times10^{-10}$$

which lies directly inside the observed BBN baryon-to-photon window $[5.8\times10^{-10},6.5\times10^{-10}]$ (Planck reference: 6.1×10^{-10}).

3D harmonic stage K	Volumetric e-folds $3K$	Scale factor e^{-3K}	Derived asymmetry $\eta_K=\tanh((\pi)/(4)e^{-3K})$	Physical regime
K=1	*3*	*4.9787× 10^{-2}*	*3.9083× 10^{-2}*	Macroscopic filament horizon floor
K=3	*9*	*1.2341× 10^{-4}*	*9.6926× 10^{-5}*	Intermediate vortex cascade
K=5	*15*	*3.0590× 10^{-7}*	*2.4026× 10^{-7}*	Pre-freeze-out cascade
K=6	*18*	*1.5230× 10^{-8}*	*1.1962× 10^{-8}*	Pre-baryonic harmonic
K=7	*21*	*7.5826× 10^{-10}*	*5.955329× 10^{-10}*	**Observed BBN window** *[5.8,6.5]× 10^{-10}*
K=8	*24*	*3.7751× 10^{-11}*	*2.9650× 10^{-11}*	Sub-baryonic harmonic

9.5 Theorem 5: Universal Exponential-Wall Bypasses, the `(1, 1, 1)` Orthogonal Triad Winner, and Hard-Core Nuclear Repulsion

Whenever compression or winding in a lower-dimensional or single-signed regime hits an exponential or superlinear resistance wall, **A0 (Path of Least Resistance)** proves that nature does not climb the wall; instead, the winning trajectory bypasses the wall by opening a new orthogonal degree of freedom or crossing `0` into `-` (`exponential_wall_bypasses_and_triad_winner` and `derived_three_accumulator_bubble_ladder`):

1. **Wall 1 — 1D Axial Compression Wall → Helical Buckling Plateau (`1D -> 2D/3D`):** Compressing a straight 1D line ($\omega=0$) hits the exponential wall $E_{1D}(\chi)=e^{+\chi}\to\infty$. Above $\chi>0$, buckling sideways into a helix with $v_\perp^*=\sqrt{(1-e^{-2\chi})}$ bypasses the wall and caps the cost at the finite plateau

$$E^*_{\text{helix}}(\chi) = \tfrac{3}{2} - \tfrac{1}{2}e^{-2\chi} \to 1.5.$$

2. **Wall 2 — 3D `+` Compression Wall → Signed `0`-Crossing Plateau (`+ -> 0 -> -`):** Forcing all winding to remain in the parent `+` region ($\theta=0$) hits the exponential wall $E_{stay,+}(\chi)=2e^{+\chi}+2e^{-\chi}\to\infty$. Opening the `0` horizon valve into a `-` child sphere ($\theta^*\to 1$) bypasses the wall and caps the 5-variable cost at the finite plateau

$$E^*_{5-\text{var}}(\chi) = 4 - \tfrac{9}{2}e^{-\chi} + O(e^{-2\chi}) \to 4.0.$$

3. **Wall 3 — Single-Axis Winding Wall $N^{3/2}$ → Orthogonal `(1, 1, 1)` Triad Ground State & Nuclear Hard Core:** For a vacuum bubble of radius R carrying integer winding $n\in Z$ (`vacuum_bubble_crystallization_and_mass_spectrum`), balancing surrounding-medium flow energy $E_{surr}(R)=m_0\,n^2/R$ ($r\geq R$) against cavity pressure work $W_{press}(R)=PV_0\,R^3$ yields the *3:1* Virial split (*75%* in the surrounding flow, *25%* in cavity work, *0%* in a point singularity), sharp integer burst lines ($z(1)=z(0)\Rightarrow n\in Z$), and *180°* two-ripple annihilation for $(+1,-1)\to 0$. However, piling $N\geq 2$ windings onto a single axis costs $E_{single}(N)\propto N^{3/2}>N$, a superlinear wall ($2^{3/2}\approx 2.828>2$, $3^{3/2}\approx 5.196>3$). Distributing integer windings $(n_1,n_2,n_3)\in Z_{\geq 0}^{\,3}$ ($m_i=m_0\,n_i^2$) across the 3 orthogonal axes of 3D space gives the exact **A0** minimum

$$E^*(n_1,n_2,n_3) = 4(PV_0)^{1/4} m_0^{3/4} \sqrt{n_1 n_2 n_3}$$

$$\lambda_i^* = \frac{n_i^2}{(n_1 n_2 n_3)^{1/2}}.$$

- $N=1$ `(1, 0, 0)` (1D loop) and $N=2$ `(1, 1, 0)` (2D sheet) have zero 3D volume ($V^*=0$) and collapse along their unwound axes under 3D pressure (likewise, under iterated **A1** binomial windings $W_k(n)=C(n,k)$, $n=3$ is the minimum integer with $W_1W_2W_3=9>0$ supporting a 3D volume, with untuned ratio $E_{3D}^*(3)/E_{1D}^*(1)=6$).
- $N=3$ `(1, 1, 1)` **is the unique minimum-cost 3D ground state:** $E^*(1,1,1)=4.0$, cost per winding $4/3\approx 1.3333$, and isotropic aspect ratios *1:1:1* (a round sphere).
- $N=4$ `(2, 1, 1)` **is the first excited 3D state:** Because 3D space has only 3 orthogonal axes, adding a 4th winding forces doubling on one axis to `(2, 1, 1)`, breaking spherical symmetry into a *4:1:1* prolate state with higher cost per winding $E^*(2,1,1)/4=\sqrt{2}\approx 1.4142>4/3$ (excitation gap $\Delta E=4(\sqrt{2}-1)\approx 1.6569$).
- **Medium-range attraction + short-range hard-core repulsion:** Two `(1, 1, 1)` triads separated by $d>2R^*$ attract via shared surrounding flow ($\Delta E_{flow}=-2m_0/d<0$), reaching a bound minimum at contact $d=2R^*$ ($E_{contact}=8.0-1.0=7.0$). However, merging them into a single `(2, 2, 2)` bubble at $d\to 0$ costs $E^*(2,2,2)=8\sqrt{2}\approx 11.3137>2E^*(1,1,1)=8.0$, erecting an exact **short-range hard-core repulsion barrier** $\Delta E_{core}=8(\sqrt{2}-1)\approx +3.3137$.

9.6 Complete Non-Local to Local Observer Dictionary & Quarantine of Post-Hoc Coincidences

Non-local multiverse geometry (A0--A4)	Local physics label	Derived mathematical relation & calculated readout
Circle $z(\tau)=R_0e^{i\omega\tau}$ seen edge-on as pendulum $R(\tau)=R_0cos(\omega\tau)$ crossing 0 into -	Black-hole singularity / Big Bang birth	Smooth crossing ($\Delta\dot{R}=0$, $\ddot{R}(\tau_0)=0$); unsigned size $r=\|R\|\geq 0$ creates an artificial velocity cusp $\Delta\dot{r}=+2R_0\omega$ at $r=0$
Two half-turns $(-1)\times(-1)=+1$ across two nested 0 crossings ($P^2=(-I_3)^2=+I_3$)	Parity / orientation restoration across two generations	$det\ P=(-1)^3=-1$ (*Parent (+)→Child (-)*), $det(P^2)=(-1)(-1)=+1$ (*Child (-)→Grandchild (+)*)
Exponential-wall bypasses (1D -> helix, + -> 0 -> -, single-axis $N^{3/2}$→ 3D triad (1,1,1))	3D nucleon ground state & nuclear hard-core potential	(1,1,1) ($N=3$) is the unique minimum-cost 3D ground state ($E^*=4.0$, *1:1:1* sphere); (2,1,1) ($N=4$) is *4:1:1* prolate ($E^*=4\sqrt{2}$); two (1,1,1) triads attract at $d>2R^*$ ($-2m_0/d$) and repel at $d\to 0$ against (2,2,2) (*+3.3137* hard core)
Surrounding-flow Virial lock $E_{surr}^*/W_{press}^*=3$ & integer loop closure $z(1)=z(0)$	Mass in surrounding flow, quantized line bursts & *180°* annihilation	*75%* of bubble energy in $r\geq R$ flow, *25%* in cavity, *0%* in a point; *(+1,-1)→ 0* releases two equal *180°* ripples
Isotropic 3D stretch $3(\theta^*)^{3/2}$ inside - sphere fed by outer birth 0 valve	**Dark Energy (Ω_Λ)**	$\Omega_{DE}(\chi_*)=71.1441\%$ at $\Omega_{vis}=e^{-3}$ (*70.8827%* at $\chi=3$; Virial limit *71.2660%*)
Shared 0-wall enclosing load $(\theta^*)^{3/2}$ + radial infall load $2(1-\theta^*)e^{-\chi}$ across inner/outer 0 horizons	**Dark Matter (Ω_{DM})**	$\Omega_{DM}(\chi_*)=23.8772\%$ at $\Omega_{vis}=e^{-3}$ (*23.8134%* at $\chi=3$; Virial limit *23.7553%*)
Active 1D spaghettified filaments $2(1-\theta^*)^2e^{\chi}$ pinching toward inner child 0 horizons	**Visible Matter (Ω_b, cosmic web)**	$\Omega_{vis}(\chi_*)=e^{-3}=4.9787\%$ (*5.3039%* at $\chi=3$), filament onset $\chi_{spaghetti}=(1)/(2)ln\ 2$
Single-sign orientation entry $sign(det\ F_k)=(-1)^k$ + offset decay harmonic ladder	**Missing Antimatter & Baryon Asymmetry (η_B)**	*100%* single-signed bulk interior; ladder $\eta_K=tanh((\pi)/(4)e^{-3K})$ gives $\eta_7=5.955329\times 10^{-10}$ at $K=7$
Moving room wavelength band $[\lambda_{min}(\chi),\lambda_{max}(\chi)]=e^{-\chi}[\lambda_{min,0},\lambda_{max,0}]$	**Planck Length & Absolute Zero**	Shortest wave before $0\to$ - (λ_{min}) and longest wave as wide as the room ($\lambda_{max}=2R$), shifting together across nests
Closed Navier--Stokes vortex--diffusion cycle (oo / OO oO / oO Oo / oo)	**Heat Death vs. Cyclic Renewal**	Diffused still-air o rooms ($\to\lambda_{max}$) re-accumulate into giant hurricane 0 rooms ($\to\lambda_{min}$) with zero loss

Quarantine of candidate numerical coincidences not derived from A0--A4: To maintain strict separation between proved geometric theorems and post-hoc numerical comparisons (numerical_coincidences_not_derived), the following expressions are recorded as non-derived candidate coincidences: (i) Lenz's 1951 number $6\pi^5\approx 1836.1181$ and its two-valve dressing $6\pi^5+e^{-3}/\sqrt{2}\approx 1836.1533$ (because the discrete **A0+A1** binomial ladder gives $E_{3D}^*(3)/E_{1D}^*(1)=6$ without a derived π^5 continuum factor); (ii) the threshold *n–p* gap formula $2+(1)/(2)+e^{-3}/\sqrt{2}\approx 2.5352\ m_e$; (iii) selecting rung $K=7$ on the η_K ladder without an independent freeze-out equation; (iv) CMB temperature projections using measured H_0 and G or an ad-hoc $-4/5$ exponent; and (v) the *+2.67%* difference between the two-valve e^{-3} fixed point (*71.14%* DE / *23.88%* DM) and the Planck base-ΛCDM fit (*68.47± 0.73%* DE).

9.7 One phase now drives a finite enclosing geometry, zero-radius opening, and last-echo cutoff

The coupled geometric continuation adds explicit phase and geometric inertia to A0. Available regional volumes are $V_i=R_i^3-\sum_j R_j^3>0$ and path stretches are $\lambda_i=V_i^{1/3}$, with the outer radius free. One continuous phase has kinetic weight λ_i and stretch cost $\psi_x^2/(2\lambda_i)$; geometry has the declared logarithmic-boundary inertia $3\sum_i \ln \dot{R}_i^2/2$. These kinetic and volume-to-path rules are explicit candidate assumptions. The parent pressure is now derived, $P_i=E_i/(3V_i)$, rather than supplied as a driving history. Its work and the receiving field obey

$$\dot{E}_i = F_i - E_i \ln \dot{\lambda}_i$$
$$\dot{K}_g = \sum_i E_i \ln \dot{\lambda}_i$$
$$\sum_i F_i = 0.$$

Characteristic coordinates carry the same field beyond the numerical break caused by exponentially stretched material cells. At a transparent three-curve junction, a single incoming parent wave reflects energy *1/9* and sends *4/9* into each quiet receiver. This is a new exact graph-action relation, distinct from the earlier *88.4735%* scalar-horizon calculation.

The free-parent limit has $Q_0=4\pi^2$, $a=\sqrt{Q_0}/6$, $\lambda=\cosh^2(a\tau)$, $E_\psi=Q_0/\lambda$ and $K_g=Q_0\tanh^2(a\tau)$. Its declared propagation metric has finite future optical reach $1/a=0.3898484006$, with proper causal-boundary length tending to $1/(2a)=0.1949242003$. No vacuum potential sets this expansion law.

Replacing logarithmic inertia at small radius by radial-displacement inertia admits a continuum wave-driven opening from zero radius with $r(\tau)=\pi^2\tau^2-(4)/(3)\pi^4\tau^3+O(\tau^4)$ and receiving budget $4\pi^4\tau^2+O(\tau^3)$ ($2.9718260669\ \tau^2$ in the affine reduction). The hybrid radial inertia $M(r)=1/(1+r^2/3)$ satisfies both the finite-distance birth criterion $\int_0^\varepsilon \sqrt{M(r)}\,dr<\infty$ and the finite-future optical-reach criterion $\int^\infty \sqrt{M(r)}\,dr/r<\infty$ (giving free-room optical reach 0.2611589801). In the coupled hybrid evolution, reflected echoes have a sharp launch cutoff at 0.2269423871 at the mouth and 0.1165029354 at a parent detector 0.01 material units away, with arrival delay diverging as $-\ln(\delta)/\kappa$ and returned energy vanishing linearly as $\kappa\delta$. The reference calculation gives finite chain/branch transfer and numerical limits. The receiving tree is admitted initially; a dynamically formed child and identification of a contact with the proposed one-way horizon remain to be derived.

9.8 Exact budget surface, causal neck onset, and the three-axis orthogonal cascade

Three further exact mathematical laws follow directly from the declared cost and exterior phase equations:

1. **Exact algebraic surface across the 5-variable budget curve (`exact_budget_curve_invariant`):** Writing $D=\Omega_{DE}$, $M=\Omega_{DM}$, $V=\Omega_{vis}$ and $B=M-D/3$ for the radial infall share, eliminating χ from the 5-variable stationarity equation yields the parameter-free identity

$$9DVB = 4\left(V+\frac{B}{2}\right)^3$$
$$\theta^* = \frac{D}{D+V+B/2}$$
$$e^{-\chi} = \frac{3\sqrt{\theta^* B}}{2V+B}$$

with deep-compression scaling $B/V^2 \to 16/27$. Because $D+M+V=1$, specifying any single share $V\in(0,1)$ uniquely determines $(D(V),M(V),\chi(V))$ with zero free parameters. Alongside the amplitude-area residue

$\eta_{area}=\tanh((\pi\gamma)/(2\omega))$, the damped horizon oscillator's quadratic energy ledger fixes the geometric work residue $\eta_{work}=\tanh((\pi\gamma)/(\omega))=(2\eta_{area})/(1+\eta_{area}^2)$.

2. **Spherical fragmentation, outgoing neck waveform, and whole-exterior causal onset (`split_waveform_evolution`, `split_onset_field`):** Splitting a bubble of winding N into same-sign integer daughters $\sum_i n_i=N$ at fixed pressure preserves total equilibrium surface area $\sum_i 4\pi(R_{n_i}^*)^2=4\pi(R_N^*)^2$ while releasing energy fraction $1-\sum_i(n_i/N)^{3/2}>0$ ($1-1/\sqrt{2}\approx 29.2893\%$ for a binary split). Along the two-center equipotential neck path, coupling the inner shape coordinate $q_1=c^2$ to outgoing even multipoles (waveform summary) pinches the neck at offset 3.960294 and reaches separated lobes $s=1.2$ at 4.264066. Removing the matching-sphere approximation in the quadratic onset problem over the entire exterior $r\geq 1$ yields the boundary release condition $f_r(1,t)+4f(1,t)=0$ and characteristic cubic

$$p^3-3p-3=0$$

with unstable growth rate $p_+=2.1038034027$ and decaying oscillatory transient $-1.0519017014\pm 0.5652358517\,i$.

3. **Inherited daughter preparation, three-axis orthogonal cascade $e_z\to e_x\to e_y$, and 2×2 kinetic handoff (`inherited_daughter_calculation`):** At the first split's separated endpoint $s=1.2$ (inherited daughter summary), each sibling's field induces the other daughter's multipole preparation (normalized dipole -0.07340185, quadrupole $+0.02750552$, velocity -0.05818555) without a second supplied seed. Because the exterior phase is harmonic ($\Delta\Phi=\partial_{xx}\Phi+\partial_{yy}\Phi+\partial_{zz}\Phi=0$), the directional volume-derivative trace vanishes identically at every stage:

$$\partial_{\zeta_x}v+\partial_{\zeta_y}v+\partial_{\zeta_z}v=0.$$

After the first split along e_z, axial symmetry and $\Delta\Phi=0$ give $\partial_{\zeta_x}v=\partial_{\zeta_y}v=-(1)/(2)\partial_{\zeta_z}v<0$ (normalized energy derivative -0.03151123 transverse vs. $+0.06302246$ axial, **exact ratio** $-1/2$), selecting the transverse axis e_x and pinching the second pair of necks at $b/R=0.62746042$. Once the four lobes separate in the xz-plane ($b\geq 1.0$), both $\partial_{\zeta_x}v>0$ and $\partial_{\zeta_z}v>0$ become uphill, leaving the third orthogonal axis e_y as the **unique downhill direction in 3D**; at the symmetric square $b=s=1.2$, four-fold symmetry and $\Delta\Phi=0$ fix the **exact ratio** -2 ($\partial_{\zeta_y}v=-2\partial_{\zeta_x}v=-2\partial_{\zeta_z}v=-0.11574536<0$), pinching the four planar lobes into eight 3D octant lobes at $c_y/R=0.36044932$ ($b=1.0$) and 0.34722798 ($b=s=1.2$). At the $s=1.2$, $b=0$ handoff, $\partial_{zz}\Phi_0\equiv\partial_{cc}\Phi_0$ gives the exact multipole identity $\partial_{q_2}g_{l,0}|_{q_2=0}=-(l-1)/(2)\partial_{q_1}g_{l,0}$ and a B-independent 2×2 quasistatic kinetic matrix M_{full} ($\det M_{full}=0.14003084>0$), generating nonzero initial transverse conjugate momentum $p_2(0)=+0.05157014>0$ and a positive combined potential-plus-memory drive $-\partial_{q_2}U-R_2=+0.18951641>0$.

Appendix A. Technical atlas and preserved mathematical record

The main argument follows only the outward chain. This appendix preserves the full earlier technical chapters, their equations, tables, evidence and claim markers. Section identifiers are remapped; scientific values are unchanged. The following one-paragraph summaries locate the frozen engines and controls. Their underlying records remain available in full below and in the linked owners. No numerical campaign is resumed by this move.

A.1. GH evolution engine

The compact nonspherical Einstein/two-scalar GH implementation, exact frame derivatives, constraint diagnostics and local operator controls remain as implemented in `17098b3c`. It evolves the declared action; its existence does not select a cosmic preparation or prove a child. The full technical development and owner are retained below.

A.2. Conditioning

Metric-dependent lapse feedback and the proper-duration comparisons in `487a0b1f` control coordinate conditioning without adding physical stress. They remove an imposed coordinate-time ceiling, not an invariant singularity or a missing vacuum law. The original equations and evidence are retained in the GH owner.

A.3. Observer readouts

The proper-clock, phase and local stress readouts in `487a0b1f` and `2ed1aea0` distinguish lapse/shift coordinates from receiver measurements. Their finite apparatus protocols and supplied initial state are preserved. They do not identify the receiver event with today's cosmological epoch; the comparison retains the numerical values and limitations.

A.4. Boson transport

The canonical projection and symplectic matter-probe transport in `771c5a78`, `292eb146` and `61124309` propagate perturbations on the supplied parent geometry. Reference response norms are not energy fractions; incoming covariance and omitted metric fluctuations remain explicit. The boson owner and full original section retain the finite projections, state assumptions and records.

A.5. Dense history

The retained DOP853 dense-polynomial history in `771c5a78` supplies source-compatible metric/field samples for probe evolution without replacing the nonlinear solution by sparse interpolation or extrapolation. The history record keeps its producer and regeneration route. This computational representation does not add a new source history or a physical epoch-selection rule.

A.6. Schur memory

The retarded finite-sector reduction in `61124309` derives the memory kernel from the same canonical block evolution. Its noise requires an incoming covariance, and initial cross-correlations remain physical data. A memory kernel alone is not a vacuum stress. The owner and retained equations below preserve that distinction and the original numbers; no additional orders or precision are requested here.

A.7. Kick and shear

The York-constrained momentum preparations and amplitude-independent shear/internal-coupling relation in `b69dda73` remain finite conditional results. Their source profiles, zero freely supplied transverse-traceless datum and vacuum-reference curvature are explicit assumptions. The kick owner preserves the full derivation and tests. They do not select a present vacuum fraction or replace the missing receiving stress history.

A.8. Epsilon=.25 controls

The three-resolution control in `2ed1aea0` preserves the original nonspherical preparation and fixed proper endpoint, with decreasing constraint and adjacent readout differences. Its spreads are empirical indicators, not certified physical errors. The saved comparison is retained below unchanged. The separate epsilon=1 nonlinear runs are **under-resolved, not used for claims**; they establish no trapped threshold.

A.9. Matter-work ledgers

The perturbative balance in `6561d8cf`, passive nonlinear ledger in `e6f34eb7`, and finite controls in `2ed1aea0` retain pressure, geometry, measure and boundary contributions rather than identifying energy loss with vacuum production. Their residuals remain numerical controls in the coupled-response and GH owners. The full `v0.26.0` technical atlas and all historical chapters are preserved in the archived foundation volume and in the repository Markdown source that follows.